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1. Introduction

The fundamental element of the INS is the **Inertial Sensor System (ISS)** — a stable platform consisting of high-quality gyros and accelerometers plus a computer.

What the computer does:

- Integrates accelerometer outputs with time → velocity
- Integrates velocity with time → distance travelled
- Derives pitch/roll attitude, true heading, true track, drift, present position (lat/long), ground speed, and wind

A further navigation computer injects and stores **waypoints**, then computes track angle error, distance and time-to-go. This output can drive the autopilot, flight director, or manual flying.

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The modern INS was the **first self-contained single source** of all navigation data. The INS has now been joined by the similar IRS laser-gyro system (Chapter 19).

2. Basic Principles of INS

Newton's Laws underpinning INS:

- A body continues in rest or uniform straight-line motion unless acted on by external force.
- Acceleration ∝ force / mass.
- Every action has an equal and opposite reaction.

Einstein (1905) pointed out that "at rest" simply means moving at the same velocity as the observer. The accelerometer — the primary measuring device — makes **no distinction between rest and any fixed velocity**. It does, however, distinguish between a truly fixed velocity and one that is fixed along a curved path.

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3. Accelerometers and Integrators

Two accelerometers are mounted at the heart of the inertial system:

- One measures acceleration in the **North–South direction**
- One measures acceleration in the **East–West direction**
- A third is often fitted for **vertical acceleration**

Accelerometer operation (pendulous device):

1. Aircraft accelerates → pendulum swings off null due to inertia
2. Signal pick-off detects displacement
3. Signal → amplifier → torque motor → restores pendulum to null
4. Current into torquer = measure of acceleration

Double Integration Process

```
graph LR
  A["Acceleration
(ft/s²)"] -->|"× time
1st integrator"| B["Velocity
(ft/s)"]
  B -->|"× time
2nd integrator"| C["Distance
(ft or NM)"]
```

Position computation: The computer knows the take-off lat/long. It continuously adds N/S distance and E/W distance to compute **present position to the nearest tenth of a minute of arc**.

4. Gravity Effects on Accelerometer

If the accelerometer is tilted (not kept earth-horizontal), the pendulum is displaced by **gravity** even when there is no real acceleration. This would produce a **false acceleration signal** → false velocity → false distance.

Solution: The accelerometer must always be kept **earth-horizontal**. This is achieved by mounting it on a gyro-stabilised platform (gimbal assembly).

5. The Integrating Gyroscope

The INS uses a **rate-integrating gyro** — a one degree-of-freedom gyro using **viscous restraint** (not mechanical/spring restraint as in a rate gyro).

Construction: a can-within-a-can; the outer frame is filled with viscous fluid that supports the inner gimbal, reducing bearing torques. The inner gimbal is pivoted about its vertical axis.

6. The Platform

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The platform is a gimbal assembly that allows the aircraft to go through any attitude change while the inner element (carrying the accelerometers and gyros) remains **earth-level**.

How the platform maintains level:

1. Platform tips → gyro spin axis stays fixed in space
2. Signal pick-off detects case displacement
3. Signal → amplifier → gimbal drive motor → restores level

Three integrating gyros are mounted with **mutually perpendicular input axes**. Three gimbal motors drive the platform about pitch, roll, and vertical axes respectively.

7. Earth Orientation and Apparent Wander

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A gyro-stabilised platform remains fixed in *space* — but the aircraft operates on a *rotating, curved Earth*. Two compensations are applied by torquing the gyros:

Earth Rate Compensation (apparent wander due to earth rotation):

- Compensates for the **horizontal component of earth rate** felt by the gyros
- Varies with latitude: **zero at the equator**, maximum **±15.04°/hr at the poles**

Transport Rate Compensation (wander due to aircraft movement over earth):

- Developed using velocity signal
- Signal = aircraft velocity ÷ earth's radius

Additional compensations for **Coriolis and centrifugal effects** are also applied.

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8. Alignment of the System

The stable element must be precisely aligned in **azimuth** and **attitude** before navigation. The alignment sequence is:

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1. **Warm-up** — fluid-filled components reach operating temperature (~3–4 min)
2. **Coarse levelling** — pitch and roll driven to 90° to each other; platform roughly levelled using aircraft frame or gravity switches/horizontal accelerometers
3. **Coarse azimuth alignment** — platform turned until heading output agrees with best known True Heading. Platform aligned to within 1°–2° in seconds.
4. **Fine levelling** — zero output from accelerometers; levels platform to within **6 seconds of arc** (takes up to 1–1½ min)
5. **Gyro compassing** — east gyro detects earth rotation component when misaligned; resultant signal torques the azimuth gyro until table aligns to **True North**

Requirements before entering NAV mode:

- Accelerometers levelled (velocity set to zero)
- Platform orientated to True North (gyro compassing complete)
- Initial position (lat/long) entered accurately — **aircraft must be stationary**

9. Schuler Period

Schuler postulated an earth pendulum with length equal to the **radius of the earth** — its bob at earth's centre, suspension at the surface. If accelerated around the earth, the bob remains vertically below the suspension point (at earth's centre of gravity), so a platform tangent to the surface stays horizontal regardless of acceleration.

Schuler Tuning:

- INS stable element maintained normal to local vertical by feeding back aircraft radial velocity (V/R) as levelling gyro signals
- If the platform is displaced from horizontal, it oscillates with a period of **84.4 minutes** — the **Schuler Period**
- The INS is "Schuler tuned" — an analogue of the 84.4-min earth pendulum

10. Errors of INS

Bounded Errors

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Errors that build up to a maximum and return to zero within the **84.4-minute Schuler cycle**. Causes:

- Platform tilt due to initial misalignment
- Inaccurate measurement of acceleration by accelerometers
- Integrator errors in the **first stage** of integration

Unbounded Errors

Cumulative errors that grow with time. Causes:

- **Initial azimuth misalignment** of the platform
- **Wander of the azimuth gyro**
- Wander in levelling gyros (causes Schuler oscillation but mean distance run diverges from true)
- Integrator errors in the **second stage** of integration

Inherent Errors

Due to the **irregular shape and composition of the Earth**, movement of the Earth through space, and other physical factors. These vary from system to system based on the balance between accuracy, simplicity, reliability, and cost.

11. INS Control and Display Panels

The traditional INS uses two panels:

- **Mode Selector Unit (MSU)** — selects operating mode
- **Control and Display Unit (CDU)** — waypoint entry and data readout

Mode Selector Unit — Modes

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Mode	Function
STANDBY	Power supplied to all parts. Ramp position (lat/long to nearest 0.1') entered here.
ALIGN	Platform levelled and gyro-compassed. READY NAV illuminates when complete.
NAV	Full navigation computing. Aircraft may taxi without degrading accuracy.
ATT REF	Computing disconnected; alignment lost. Accelerometers act as gravity switches; gyros become gravity-tied (earth gyros). Gives attitude and limited heading (DGI mode). Heading must be reset periodically to magnetic source.

Battery Operation: If aircraft electrical supply fails, INS automatically switches to internal battery. BATT light illuminates on CDU. When battery power is nearly exhausted, BATT WARNING light on MSU illuminates. The INS **cannot be re-levelled or re-aligned in flight** — the aircraft must be stationary with known exact position.

12. Warning Lights Summary

Light	Indication	Action Required
READY NAV (MSU) — Green	Alignment complete	Select 'NAV'
BATT (MSU) — Red	Battery power too low for operation	Check power supplies
ALERT (CDU) — Amber	Approaching (or overflying in MAN mode) a waypoint	None, unless MAN mode — initiate TK CHG
BATT (CDU) — Amber	INS operating on back-up power	Check power supplies
WARN (CDU) — Flashing Red	System malfunction	Set selector to DSR TK/STS; note action code; consult user's guide

ALERT light behaviour:

- **AUTO mode:** Illuminates steadily **2 minutes before** waypoint; extinguishes as track changes overhead the waypoint.
- **MANUAL mode:** Illuminates steadily 2 minutes before waypoint; flashes from **30 seconds before**; continues flashing until track is changed by operator.
- Does NOT illuminate below a set speed (typically 100 kt or 250 kt).

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13. LED Display Functions (CDU Function Selector)

Function	LH Window	RH Window
TK/GS	Track (°T) to 0.1°	Ground speed (kt)
HDG/DA	True heading (°T) to 0.1°	Drift angle (L/R) to 0.1°
XTK/TKE	Cross-track distance (L/R NM) to 0.1 NM	Track angle error (L/R) to 0.1°
POS	Present latitude to 0.1'	Present longitude to 0.1'
WPT	Waypoint latitude to 0.1'	Waypoint longitude to 0.1'
DIS/TIME	Distance to next WPT (NM)	Time to next WPT (to 0.1 min)
WIND	Wind direction (°T)	Wind speed (kt)
DSR TK/STS	Desired track (°T) to 0.1°	Status (blank in NAV mode)
TEST	All digits illuminated for display check	

Waypoint 0: Represents the aircraft's position at the last time a track change from present position to a specified waypoint was selected. Used to fly direct to any waypoint from current position. Waypoint 0 will NOT accept operator-entered coordinates — it is reserved for the computer.

14. Manual and Automatic System Checks

Initial position entry (ramp position):

- Latitude error → platform will not remain earth-horizontal in NAV mode (torque motors apply incorrect rate based on wrong latitude). Gross error detected by WARN light.
- Longitude error → platform remains stable but track/distance from departure to first waypoint is wrong; all subsequent longitudes are in error by the initial input error.

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Two pre-flight waypoint checks:

1. Recall each waypoint from store onto the LED and visually **recheck lat/long**
2. Call up **DIS/TIME** and **DSR TK/STS** between consecutive waypoints and compare against the flight plan

The INS navigates very accurately between waypoints but **cannot detect operator ("finger trouble") errors**.

E/W Integration and Longitude Update

The E/W accelerometer output is integrated twice: first to E/W speed (kt), then to E/W distance (departure in NM). To convert departure to change of longitude:

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d'long (min) = departure × sec(latitude)
Equivalently: departure must be **multiplied by the secant of present latitude** to obtain d'long.

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Practice Questions & Detailed Answers

Ghost Aviator Instrumentation — Chapter 18 Questions. DGCA CPL/ATPL.

Q1. INS errors are classified as "bounded errors" and "unbounded errors". Which statement is correct?

a. An "unbounded error" increases with time; example: distance gone error due to a ground speed error

b. An "unbounded error" increases with time; example: increasing ground speed error due to platform not being levelled correctly

c. A "bounded error" is subject to sudden unpredictable random changes, most notable during pitching manoeuvres

d. A "bounded error" is "tied" to the real wander rates of the gyros on the platform

✅ Correct Answer: **A**

Unbounded errors are cumulative errors that grow with time. A distance gone error due to a ground speed error (e.g., from azimuth misalignment or levelling gyro wander) keeps accumulating — it never "comes back" within an 84.4-min cycle. Bounded errors are those that oscillate and return to zero within the 84.4-minute Schuler cycle.

Why the others are wrong:

- **B:** Platform not being levelled produces a *bounded* error (platform tilt causes Schuler oscillation that reverses). Ground speed error due to this is bounded, not unbounded.
- **C:** Bounded errors are not random — they follow the 84.4-min oscillation. Random changes would be inherent errors.
- **D:** Real wander of the levelling gyros causes unbounded distance error, not a bounded one that is "tied" to them.

Q2. Two checks for correctly entered sequential waypoints are:

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a. Select DSR.TK/STS and check status <4; select DIS/TIME and check time agrees with flight plan

b. Select DIS/TIME and check distance agrees with flight plan; then check time agrees with flight plan

c. Select DIS/TIME and check distance agrees with flight plan; select DSR.TK/STS and check track agrees with flight plan

d. Select DIS/TIME and check distance agrees with flight plan; select HDG/DA and check heading agrees

✅ Correct Answer: **C**

The two standard cross-checks are: (1) select DIS/TIME and verify distance matches the flight plan leg distance, and (2) select DSR TK/STS and verify the desired great-circle track matches the flight plan. This confirms both the correct waypoints AND correct lat/long entries.

Why the others are wrong:

- **A:** Status code <4 is not a defined check procedure described here.
- **B:** Checking only distance and time is insufficient — two legs with same distance/time could have different tracks.
- **D:** Heading is not the same as desired track (drift exists). HDG/DA is not used for waypoint verification.

Q3. In an INS the E/W accelerations are converted into E/W speed at the first stage of integration, and into E/W distance (departure) at the second stage. To convert departure to d'long (min) for longitude update:

a. Departure × cosine of present latitude

b. Direct d'long (min) without conversion

c. Departure × secant of present latitude

d. Departure × sine of present latitude

✅ Correct Answer: **C**

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Departure (NM) = d'long (min) × cos(lat). Therefore d'long (min) = departure ÷ cos(lat) = departure × sec(lat). At the equator (lat=0), cos=1 so departure = d'long exactly. At higher latitudes, the meridians converge, so more departure is needed for the same change in longitude.

Memory hook: "Departure × SEC(lat) = d'long". The word "secant" has to "expand" the longitude value because degrees of longitude get shorter as you move from the equator toward the poles.

Q4. At the second stage of integration, E/W speed is converted into E/W distance (departure). To convert departure into change of longitude it must:

a. Be divided by secant of the latitude

b. Be multiplied by secant of the latitude

c. Be divided by tangent of the latitude

d. Be multiplied by cosine of the latitude

✅ Correct Answer: **B**

Same principle as Q3: d'long = departure × sec(lat). The answer here is stated directly: multiply by secant of latitude.

Q5. The amber ALERT light on an INS control and display unit:

a. Illuminates steadily 2 minutes, in AUTO mode, before reaching the next waypoint

b. Starts flashing 2 minutes before reaching the next waypoint and goes out at 30 seconds to run

c. Illuminates if power from the aircraft bus bar has been lost and the system is on standby battery

d. Illuminates steadily after passing a waypoint in manual mode, until the next leg is programmed in

✅ Correct Answer: **A**

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In AUTO mode, the ALERT light illuminates *steadily* at 2 minutes before the waypoint and extinguishes as the track automatically changes overhead the waypoint. There is no flashing in AUTO mode.

Why the others are wrong:

- **B:** Reverses the AUTO and MANUAL behaviours — it is in MANUAL mode that the light flashes (at 30 seconds to run), not AUTO.
- **C:** Battery operation is indicated by BATT light (amber), not ALERT.
- **D:** In MANUAL, the light flashes 30 seconds before the waypoint (not after passing it).

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Q6. With reference to INS, the functions of the integrators are:
(i) At second stage of integration to suppress unbounded errors (NAV mode)
(ii) At first stage of integration to convert acceleration → speed (NAV mode)
(iii) At second stage of integration to convert speed → distance gone (NAV mode)
(iv) To align the platform (level and align modes)

a. All four statements are true

b. Only (ii), (iii) and (iv)

c. Only (i), (ii) and (iii)

d. Only (ii) and (iii)

✅ Correct Answer: **D**

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Statement (ii) and (iii) are correct — these are the two integration stages: acceleration→speed, and speed→distance. Statement (i) is wrong: integrators do not suppress unbounded errors — they actually propagate them (second-stage integrator errors ARE a source of unbounded errors). Statement (iv) is wrong: the platform is aligned by gyro compassing and torquing, not by integrators.

Q7. The computer of a north-referenced INS in flight provides compensation for:

a. Aircraft manoeuvres, real wander, apparent wander, transport wander

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b. Coriolis, real wander, apparent wander, transport wander

c. Earth rotation, transport wander, coriolis

d. Transport wander, apparent wander, coriolis, magnetic variation

✅ Correct Answer: **C**

INS computers compensate for: (1) Earth rotation rate (apparent wander = earth rate compensation); (2) Transport wander (transport rate compensation = V/R); (3) Coriolis and centrifugal effects. "Real wander" is an imperfection of the physical gyro and cannot be compensated by the computer (it is a random inherent error). Magnetic variation is irrelevant — INS operates in True reference.

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Remember: INS compensates for predictable, mathematically-determinable phenomena. Real wander (from bearing imperfections) is random and unpredictable — it cannot be computed away.

Q8. During initialization of an INS the aircraft must not be moved until:

a. The ramp position has been inserted and checked

b. The platform is levelled

c. The gyros and accelerometers are in the "null" position

d. The green "READY NAV" light has been illuminated and the mode selector switch has been set to the "NAV" position

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✅ Correct Answer: **D**

The INS must complete the full alignment sequence — warm-up, coarse levelling, coarse azimuth, fine levelling, and gyro compassing — before the aircraft moves. READY NAV illuminates when this is complete. Once in NAV mode, the aircraft may taxi. Moving before READY NAV and NAV selection destroys the alignment and means the INS cannot be re-aligned in flight.